

BOTS VS MEN FOR JOB

Automation can offer many advantages- increased efficiency, convenience, ecology, etc, but it also has the potential to eliminate human jobs. Techs and economists have already been sounding the alarm about the threat of mass layoffs due to automation for several years now.

<http://dgit.in/BotVSmn>

BIO-ENHANCING YOUR BRAIN

Bioenhancement is likely to become a basic feature of human society very soon. New brain-machine interfaces will improve our memory and cognition, senses, and confer direct control over an array of semi-intelligent gadgets but it will also make us less human.

<http://dgit.in/BrainUG>

SELF-DRIVING CARS WILL KILL

You're driving down Main Street and your brakes give out. As the terror hits, a gaggle of children spills out into the road. What do you do? It's clear what any decent human being would choose to do. But would a robotic-car make the right choice?

<http://dgit.in/robocarskill>

WHAT YOUR THERAPIST DOESN'T KNOW

Imagine a surgeon or a dancer learning without someone observing their work. That's the predicament therapists are in. Big Data has transformed everything from sports to politics to education and it will soon transform mental-health treatment, too.

<http://dgit.in/therapistAI>



pany has previously stepped up to create what is the biggest supplying battery power plant in the entire world in just 90 days, generating 80MWh of power for Southern California.

Crucial to its latest achievements is the new Gigafactory near Reno, Nevada, which cost \$5 billion to be completed. Despite the staggering expense of this state-of-the-art facility, Rive has stated that not only has Tesla managed to ramp up battery production from the factory, it has also led to a reduction in expenses per kilowatt hour of battery capacity for every battery produced. This also makes Tesla's ambitious power supply plans more viable. The company plans to use numerous battery packs of small capacities in combination to generate large amounts of power. While many were skeptical regarding the viability and longevity of such a setup, particularly in commercial layout, it seems to be working as of now.

As against previous Tesla battery packs, the Powerwall 2 packs twice the power capacity of 14kWh, all the while using a body that is 30 percent smaller than its predecessor. For massive installations, Rive stated, the cost of battery storage now stands at anywhere between \$400 and \$600 per kWh of capacity. Going upward of 100MWh of power demand, Musk stated, would bring expenses down to around \$250 per kWh. While this is commercial implementation, the Powerwall 2 that Tesla has showed yesterday can also be arranged in stack configurations for larger buildings, and can deliver maximum power of 126kWh in a nine-pack combination. For reference, an average family of four in India use up to 75kWh of power every month.

Tesla lists a Powerwall 2 battery for \$5,500, and recommends that for a 3BHK household, two Powerwall 2 batteries will be required to run all appli-



ances for one full day. The cost for setting up two batteries measure up to \$13,600 including refundable deposits, amounting to almost Rs. 10,00,000 by direct conversion in India. With alternating estimated usage, this can reduce stress on traditional power resources by up to 70 percent. The investment, though, for the time being, is quite steep.

India, however, is not ready for such an infrastructure shift. The Powerwall 2 batteries translate maximum efficiency with solar-powered panels charging them, which completely negates residual cost of powering households, as is happening in Australia. Such a city power model is not existent in India, and even in developed nations as Australia, it is only just growing in number. Government-end deployment, too, may require substantial tweaking of norms and procedures in any nation where Tesla is to deploy its battery power plant.

If Australia does accept Tesla's open invitation and goes through with the monetary proceedings (that can reach close to \$100 million), the new Tesla power plant can replace its own Southern California plant as the largest in the world. The company's focus on clean energy and lesser cost of energy establishment, even with rocket fuels,



is an ongoing procedure, one which will hopefully provide dividends in our future. Overall, even if it's not entirely clear whether Tesla, and Elon Musk, will be able to pull this off or not, but if they do - that would definitely put renewable energy on the map as a good, viable and quick source of energy 

<http://dgit.in/Tesla300MWh>